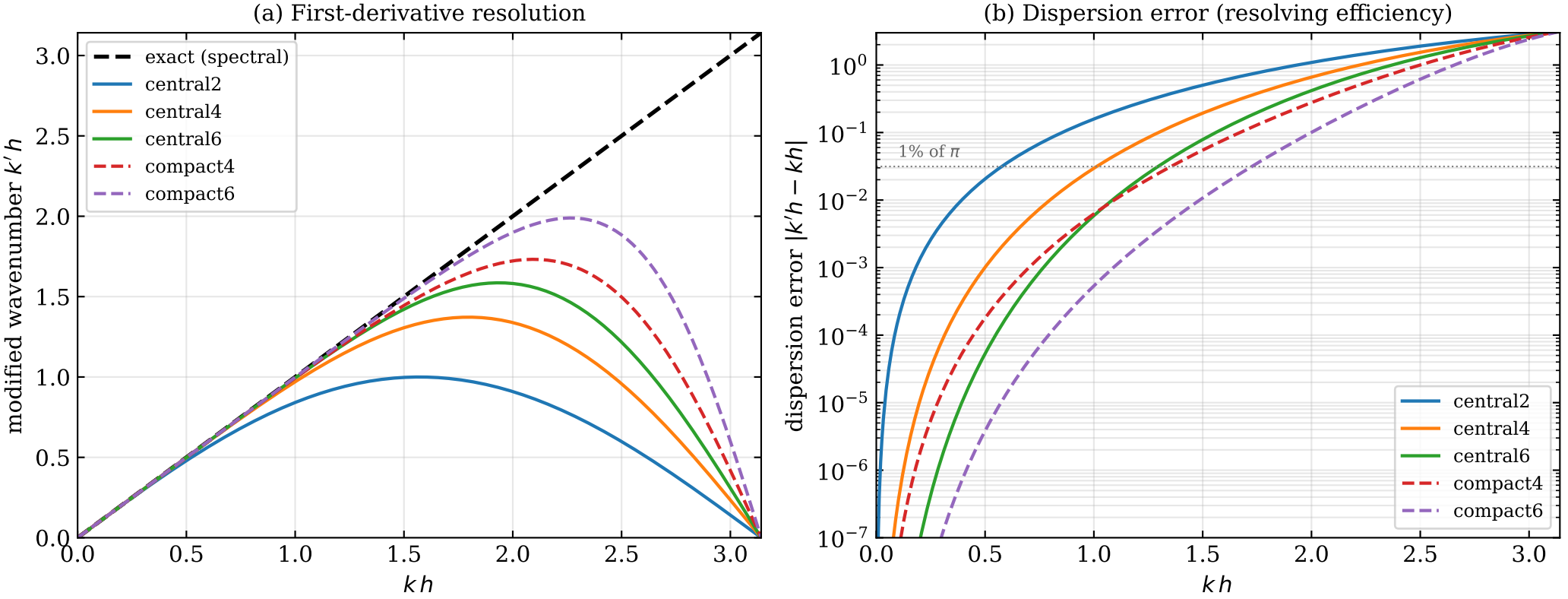
(a) First-derivative resolution
exact (spectral)
central2
central4
central6
compact4
compact6
modified wavenumber $k'h$
kh
0.0
0.5
1.0
1.5
2.0
2.5
3.0
(b) Dispersion error (resolving efficiency)
dispersion error $|k'h - kh|$
1% of π
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}